A Comparative Framework for Robust Variable Selection: Integrating Stability, Rank-Based Screening, and Adaptive Regularization

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Abstract

High-dimensional biomedical datasets, in which features vastly outnumber samples, pose a major challenge for precision medicine: identifying truly informative molecular features among thousands of candidates. Feature selection methods span diverse paradigms, including adaptive regularization, iterative screening, and stability selection, but practical performance across varying data complexity remains unclear. We conduct an empirical comparison of four methods on synthetic datasets with known ground truth and real cancer genomics data, with particular focus on SPARCS (Stable Permutative Adaptive Rank-Based Correlation Screening), the novel integrated framework and contribution of this paper, combining rank-based screening, nonlinear dependence detection via the Randomized Dependence Coefficient, permutation testing, and batch Complementary Pairs Stability Selection. We evaluated methods on two synthetic regression datasets differing in nonlinearity and complexity and two cancer gene expression datasets from the CuMiDa repository. In moderately complex synthetic data, SPARCS achieved the highest feature recovery while maintaining competitive predictive performance. Under extreme nonlinearity and noise, however, adaptive Elastic Net outperformed all screening-based methods, improving predictive accuracy as complexity increased, highlighting the robustness of strong $L_1 + L_2$ regularization against nonlinearity especially for multiclass classification. All methods achieved comparable performance on binary classification but differed substantially in multiclass performance and computational efficiency. Feature set overlap across methods was low, indicating divergent but valid selection priorities. SPARCS demonstrated adaptive sparsity, selecting a substantially smaller feature set for multiclass classification while achieving competitive accuracy. These results provide practical guidance for method selection in high-dimensional biomedical studies and highlight fundamental trade-offs between sensitivity, stability, computational efficiency, and robustness to complexity.

1 Introduction

Modern biomedical research compiles and generates datasets where the number of measured variables (features) often vastly exceeds the number of samples, a scenario known as the "high-dimensional" or "large p, small n" problem [1, 2]. For example, a gene expression study might measure 20,000 genes across only a few hundred patients [2]. In such settings, most measured features are irrelevant noise, and identifying the truly informative subset is both statistically challenging and computationally demanding [1, 3]. Selecting too many features leads to overfitting, poor generalization, and unreliable interpretability, while missing important features results in lost predictive power and scientific insight [3].

Traditional statistical methods are not applicable in high dimensions: classical regression assumes more samples than features, and standard hypothesis testing suffers from multiple testing problems when screening thousands of variables simultaneously [4]. Moreover, biomedical data in particular poses unique challenges. Features are often correlated due to biological mechanisms such as co-expression of genes in regulatory pathways [5], feature-outcome relationships may be nonlinear or involve complex interactions [6, 7], and data distributions are usually heavy-tailed with outliers due to measurement noise and biological heterogeneity [8].

1.1 Existing Approaches and Their Limitations

Feature selection literature has developed various families of methods to address these challenges, each differing in theoretical foundations and practical trade-offs [3, 9, 10].

Regularization-based methods. These perform feature selection by adding penalty terms to regression coefficients during model fitting, setting irrelevant features to exactly zero. The LASSO (Least Absolute Shrinkage and Selection Operator) uses an L_1 penalty that pushes strong sparsity [11]. The Elastic Net extends this by combining L_1 and L_2 (Ridge) penalties to handle correlated features [12]. Adaptive variants incorporate per-feature penalties based on initial estimates, improving selection consistency and demonstrating oracle properties under specific conditions [13]. These methods are computationally efficient and theoretically backed, but they assume linearity and can be sensitive to the chosen regularization strength [4]. Thus, they struggle with nonlinear relationships and may miss important features when correlations are strong.

Screening-based methods. These take a different approach by first vastly reducing the feature space through marginal correlations before applying more sophisticated selection algorithms. Sure Independence Screening (SIS) and its iterative variant (ISIS) use correlations to quickly eliminate most irrelevant features, then iteratively refine the selection [14–16]. These methods are extremely fast and provide theoretical guarantees under certain conditions. For one, they reduce computational complexity from $O(p)$ model fits to a single-pass marginal screening step. However, standard implementations use Pearson correlation, which is sensitive to outliers and assumes linear, monotonic relationships. Robust variants have been built to address these limitations, a notable one involving Spearman rank correlations, which can resist outliers and capture monotonic nonlinear relationships [17–19]. Even with these improvements, correlation-based screening measures only pairwise linear or monotonic associations, often missing features with more complex nonlinear or interaction effects.

Stability-based methods. These address a different concern: selection stability and false discovery control [20, 21]. Even if a method illustrates good predictive performance, the selected feature subset may change dramatically with small data perturbations. This is important for scientific interpretability and clinical translation: a signature that varies wildly across patient cohorts lacks biological meaning and practical utility [21, 23, 24]. Stability selection, introduced by Meinshausen and Bühlmann, repeatedly applies a base selector to random dataset subsamples and only retains features selected consistently across subsamples according to a chosen stability threshold [20]. Complementary Pairs Stability Selection (CPSS), developed by Shah and Samworth, enhances this process with finite-sample error control bounds [22]. This framework provides rigorous statistical control that most other methods lack. However, stability selection is computationally expensive, requiring many repeated applications of the base selector, and tends to be conservative, often selecting fewer features than optimal for prediction [25].

1.2 Integration and Nonlinear Dependence

More recent methodological developments have been exploring ways to integrate the strengths of different tools while balancing their individual limitations. One particularly promising direction involves incorporating measures of nonlinear dependence into the screening process. The Randomized Dependence Coefficient (RDC), introduced by Lopez-Paz and colleagues, provides a nonparametric measure that captures nonlinear, non-monotonic relationships missed by correlation [26]. RDC works by transforming features to copula space, applying random projections through trigonometric feature maps, and subsequently computing the canonical correlation of those projections. Importantly, RDC can detect relationships like quadratic effects, threshold phenomena, and complex interaction patterns that standard correlation entirely misses [26].

Integrating RDC into iterative screening frameworks opens doors to combining computational efficiency with successfully capturing complex biological relationships. Pairing RDC with permutation-based statistical calibration can provide nonparametric p-values without distributional assumptions [27]. When further combined with batch stability selection [25] and adaptive threshold mechanisms [28], such hybrid approaches offer a promising path towards robust, powerful, and computationally practical feature selection for high-dimensional biomedical data.

1.3 Empirical Validation and Benchmarking

Despite extensive theoretical work on feature selection methods, empirical comparisons on realistic biomedical data remain surprisingly limited and often incomplete [2, 9, 24]. Many studies evaluate methods on only one or two datasets, focus exclusively on predictive performance while ignoring selection stability, or fail to compare across different methodological families. This is especially problematic in cancer genomics, where the heterogeneity of biological mechanisms and data collection protocols means that method performance can vary substantially across applications [2, 28, 29].

To address these challenges, rigorous benchmarking studies require both synthetic datasets with known ground truth (enabling direct assessment of feature recovery) and real datasets reflecting practical applications (assessing real-world performance). The Curated Microarray Database (CuMiDa) repository provides an excellent resource for such benchmarking [29]. Feltes and colleagues manually curated 78 cancer microarray datasets from over 30,000 studies in the Gene Expression Omnibus, applying rigorous filtering, background correction, normalization, and quality control. Each dataset was carefully processed to remove technical artifacts while preserving biological signals, making them ideal for machine learning algorithm comparison [29].

1.4 Research Question and Contributions

This study addresses the following question: how do modern feature selection paradigms, namely adaptive regularization, iterative rank-based screening, stability selection, and integrated hybrid approaches, compare in their ability to identify truly informative features and achieve strong predictive performance on high-dimensional biomedical data?

To answer this question, we implemented and systematically compared four distinct feature selection methods representing distinct paradigms, the first of which is a novel contribution of this paper.

1. **SPARCS** (Stable Permutative Adaptive Rank-Based Correlation Screening). A novel integrated framework combining rank-based initial screening for robustness, RDC-based nonlinear dependence detection, permutation-based statistical testing for rigorous calibration, batch CPSS for stability without excessive computation, and adaptive mechanisms for efficiency.

2. **Iterative Sure Independence Screening with Rank-Based Spearman Correlations** (Spearman-ISIS). Iterative greedy screening methodology using robust rank correlations for marginal screening and conditional selection [15, 17, 18]. This serves as a baseline for modern screening approaches.
3. **Complementary Pairs Stability Selection** (CPSS-Spearman). Pure stability selection methodology, wrapping Spearman-ISIS with Complementary Pairs Stability Selection [20, 22]. This provides conservative selections with formal error bounds.
4. **Adaptive Stochastic Gradient Descent Elastic Net** (ASGDR for regression, ASGDC for classification). Efficient adaptive regularization using feature-specific penalties based on initial estimates [12, 13]. This combines the oracle properties of adaptive methods with computational efficiency through SGD optimization.

Each method was evaluated using a consistent framework: all methods perform feature selection on training data, then the selected features are used to train a LightGBM model with hyperparameter tuning [30]. This ensures fair comparison, as all methods are judged by the same powerful downstream predictor, isolating the quality of feature selection itself.

2 Methods and Framework

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Figure 1: The comparative feature selection pipeline. Each method receives the same training split, and every selected subset is passed to the same tuned LightGBM predictor.

2.1 Problem Formulation and Notation

Consider a dataset with n samples and p features, where $p \gg n$ (the high-dimensional setting). Let $X \in \mathbb{R}^{n \times p}$ denote the feature matrix, where each row represents a sample (e.g., a patient) and each column represents a feature (e.g., a gene's expression). Let $\mathbf{y} \in \mathbb{R}^n$ denote the response vector (continuous for regression, categorical for classification).

The feature selection problem is to identify a subset $S \subseteq \{1, 2, \ldots, p\}$ of relevant features such that S contains all (or realistically most) truly informative features (high sensitivity/recall); S excludes all or most noise features (high specificity/precision); and a predictive model trained on X_S achieves strong generalization on test data.

Let $S^\star$ denote the true set of informative features. For synthetic data where $S^\star$ is known, we evaluate

$$\text{TPR} = \frac{|S \cap S^\star|}{|S^\star|}, \qquad \text{FDR} = \frac{|S \setminus S^\star|}{|S|}, \qquad \text{Precision} = \frac{|S \cap S^\star|}{|S|}. \tag{1}$$

For real data (unknown $S^\star$), we evaluate indirectly through predictive performance on held-out test sets.

2.2 Feature Selection Methods

2.2.1 SPARCS

SPARCS is a novel integrated framework combining multiple robust techniques meant for the high-dimensional, low-sample-size biomedical data setting. It integrates robust rank-based marginal screening to account for outliers and heavy-tailed distributions, nonlinear dependence detection to capture complex feature-outcome relationships, permutation-based statistical calibration to control false discoveries, stability selection to balance sensitivity, specificity, and computational efficiency, and adaptive pool shrinking. Throughout, the internal working model is the Adaptive Elastic Net of Section 2.2.4: it fits the current selected set, supplies the residual that drives the next screening round, and provides the coefficients that stability selection counts. The algorithm proceeds as follows.

Step 1: Rank-based Sure Independence Screening warm start. For each feature j, rank-transform $X_{:,j}$ and $\mathbf{y}$ and compute the Spearman correlation $\rho_j = \mathrm{corr}(R_{X[:,j]}, R_{\mathbf{y}})$. The working set is initialized with the top d features by $|\rho_j|$, where $d = \lceil n/\log n \rceil$ following the standard SIS rate [14]. This warm start supplies the first residual; the iterative screening in Steps 2–8 continues to operate over all $p - |S_t|$ unselected features rather than being confined to the initial d.

Step 2: Iterative residual-based refinement. At iteration t with current selected set S_t, fit the Adaptive Elastic Net on X_{S_t} to obtain predictions $\hat{\mathbf{y}}$ and form the working residual $\mathbf{r}$. For regression $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$; for binary classification $\mathbf{r} = \mathbf{y} - \hat{P}(y = 1 \mid X)$; for multiclass classification $r_i = 1 - \hat{P}(y = y_i \mid x_i)$, the residual probability mass on the true class.

Step 3: Staged RDC screening. Computing the Randomized Dependence Coefficient against every one of the $p - |S_t|$ remaining features at every iteration dominates the cost of the method, so screening is staged. Stage 1 ranks the remaining features by $|\rho_j|$ against $\mathbf{r}$ and retains the top M_{pre} (default 500). Stage 2 computes

$$\mathrm{RDC}(X_{:,j}, \mathbf{r}) \in [0, 1] \tag{2}$$

for each survivor and retains the top M_{rdc} (default 200, shrunk per Step 8). RDC transforms both marginals to the empirical copula, applies k random sine and cosine projections with bandwidth s, whitens each projected space, and returns the largest canonical correlation between them [26]. We use $k = 20$ and $s = 1$, which keeps the statistic near zero under independence while recovering quadratic, saturating, and threshold dependence that rank correlation misses. The whitening step is what makes the returned quantity a correlation on $[0, 1]$ rather than a scale-dependent covariance norm.

Step 4: Permutation calibration. For the M_{rdc} surviving candidates, permute the residual n_{perm} times (default 30), recompute RDC on each permuted replicate, and form the empirical p-value

$$p_{\mathrm{emp}} = \frac{1 + \#\{\text{permuted RDC} \geq \text{observed RDC}\}}{1 + n_{\mathrm{perm}}}. \tag{3}$$

For classification the permutation is stratified within class, so the null preserves the class structure and only breaks the feature-residual association. Note that the attainable minimum is $1/(1+n_{\mathrm{perm}})$, so n_{perm} must satisfy $1/(1 + n_{\mathrm{perm}}) \leq \alpha$; at $\alpha = 0.05$ this requires $n_{\mathrm{perm}} \geq 19$, which the default of

30 satisfies. Candidates with $p_{\text{emp}} \leq \alpha$ are retained. A hybrid rule additionally retains the $m_{\min}$ (default 10) strongest candidates by RDC regardless of significance, which guarantees the iteration has something to score when the permutation null is not separable at the chosen level; the stability and magnitude filters of Steps 6 and 7 remain in force, so this floor widens the candidate pool without weakening the admission criteria.

Step 5: Batch CPSS stability. For B complementary subsample pairs (default 30), randomly split the data into complementary halves of size $n/2$ and fit the working model once per half on the full candidate set. A candidate is credited when its coefficient magnitude exceeds 10^{-8}. The stability score of feature j is the proportion of the $2B$ half-samples that select it. Because every candidate is evaluated inside a single fit rather than one fit per candidate, the batch scheme costs $O(2B)$ model fits instead of $O(B \cdot |\text{candidates}|)$.

Step 6: Adaptive stability threshold. Sort the stability scores in decreasing order, lightly smooth the profile, and take the second difference to locate the point of maximum curvature [28]. The threshold is set at that elbow and floored at $\tau = 0.6$, so the data-driven threshold can only ever be more conservative than the fixed one.

Step 7: Multi-add. Select the top k_{add} stable candidates (default 3) that clear both the adaptive stability threshold and a minimum RDC magnitude. The RDC threshold is 0.2 for synthetic regression data and 0.1 for the real classification datasets.

Step 8: Adaptive pool shrinkage. The cap on the candidate pool shrinks geometrically,

$$K_{\text{cp}}(t) = \max\bigl(K_{\min},\ \lfloor K_{\text{init}} \cdot \gamma^{t-1} \rfloor\bigr), \tag{4}$$

with shrinkage rate $\gamma = 0.9$ and floor $K_{\min} = 50$. Later iterations screen a narrower pool, on the reasoning that the strongest residual associations surface early.

Step 9: Stopping criteria. Iteration halts when (i) no candidate clears both thresholds, (ii) the feature budget $|S| =$ `max_features` is reached, or (iii) the candidate pool is exhausted. Criterion (i) is what produces data-driven sparsity: SPARCS stops on its own rather than padding the set out to a fixed target.

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Figure 2: The complete SPARCS algorithm. Steps 2–8 repeat until one of the three stopping criteria fires.

2.2.2 Spearman-ISIS

Spearman-ISIS adapts Iterative Sure Independence Screening to use Spearman rank correlations for robustness to outliers and monotonic nonlinearities [14–16].

1. Initialize $S_0 = \emptyset$ and $\mathbf{r}_0 = \mathbf{y} - \bar{\mathbf{y}}$.
2. For $t = 1, 2, \ldots$:
   (a) For each $j \notin S_t$, compute $\rho_j = \text{spearman}(X_{:,j}, \mathbf{r}_{t-1})$.
   (b) Select the top $d = \lceil n/\log n \rceil$ candidates by $|\rho_j|$.
   (c) Fit the Adaptive Elastic Net on $S_{t-1} \cup \{\text{candidates}\}$.
   (d) Greedily commit the single candidate with the largest coefficient magnitude to form S_t.
   (e) Refit on S_t and update the residual $\mathbf{r}_t$.
   (f) Stop when the largest candidate magnitude falls below 10^{-6}, the feature budget is reached, or all features have been considered.

Key characteristics: iterative residual-based screening, greedy one-per-iteration selection, rank-based robustness, and computational efficiency.

2.2.3 CPSS-Spearman

CPSS-Spearman applies pure Complementary Pairs Stability Selection using Spearman-ISIS as the base selector [20, 22].

1. For $b = 1, \ldots, B$ pairs (default $B = 30$):
   (a) Randomly partition the data into disjoint halves A_b and B_b of size $n/2$.
   (b) Apply the base selector to each half to obtain $S_{A_b} = \text{Spearman-ISIS}(X_{A_b,:}, \mathbf{y}_{A_b})$ and $S_{B_b} = \text{Spearman-ISIS}(X_{B_b,:}, \mathbf{y}_{B_b})$.
   (c) Feature j is counted in pair b if $j \in S_{A_b}$ or $j \in S_{B_b}$.
2. Aggregate selection frequencies $\pi_j = \frac{1}{B}\sum_{b=1}^{B} \mathbb{I}\{j \text{ selected in pair } b\}$.
3. Return $S = \{j : \pi_j \geq \tau\}$, with stability threshold $\tau = 0.6$ by default.

Error control. Let $\hat{q}$ be the average number of features selected per half-subsample. For $\tau > 0.5$ [22],

$$\mathbb{E}[\#\text{false positives in } S] \leq \frac{\hat{q}^{\,2}}{(2\tau - 1)\, p}, \tag{5}$$

a conservative, distribution-free, finite-sample bound requiring no asymptotic assumptions. The accompanying implementation reports $\hat{q}$ and the resulting bound alongside every CPSS run.

2.2.4 ASGD (Adaptive Elastic Net)

ASGD implements the Adaptive Elastic Net trained efficiently via stochastic gradient descent [11–13], with objective

$$\min_{\beta} \; \frac{1}{2n}\|\mathbf{y} - X\beta\|_2^2 + \lambda \left[(1-\alpha)\|\beta\|_2^2 + \alpha \sum_j w_j|\beta_j|\right], \qquad w_j = \frac{1}{\left(\left|\hat{\beta}_j^{\text{init}}\right| + \epsilon\right)^{\gamma}}, \tag{6}$$

where $\hat{\beta}^{\text{init}}$ are initial ridge estimates, $\gamma = 1$ is the adaptive exponent, and $\epsilon = 10^{-8}$. Larger initial estimates receive smaller penalties, improving selection consistency with oracle properties.

Implementation through rescaling. The weighted penalty is removed by a change of variables, which lets a standard fast Elastic Net solver do the work.

1. Fit the initial model to obtain $\hat{\beta}^{\text{init}}$ and compute the weights w_j. For regression this is a ridge fit with $\alpha = 0.1$; for classification it is an L_2-penalized logistic model with the same strength, and for multiclass problems the per-feature initial magnitude is the norm of the coefficient vector across classes.

2. Compute rescaling factors $s_j = 1/w_j$.
3. Rescale the feature matrix, $\tilde{X}_{:,j} = X_{:,j}/w_j$.
4. Fit a standard Elastic Net on $\tilde{X}$ using stochastic gradient descent.
5. Recover coefficients $\beta_j = \tilde{\beta}_j/w_j$.

This transformation is exact: $X\beta = \tilde{X}\tilde{\beta}$ leaves the fitted values unchanged, and $\sum_j w_j|\beta_j| = \sum_j |\tilde{\beta}_j|$ turns the weighted L_1 term into a plain L_1 term, so the rescaled problem a standard solver sees is the adaptive problem we intended to solve. For feature selection, features are ranked by adaptive importance,

$$\text{Importance}_j = \frac{|\beta_j|}{w_j}, \tag{7}$$

and the top k are returned, with k set to the same feature budget given to the other methods.

2.3 Datasets

2.3.1 Synthetic Regression Datasets

Two synthetic datasets (s1, s2) with continuous targets were generated. Each contains 10,000 gene-like features, 500 patient-like samples, and 80 truly informative features providing known ground truth. Features are generated in equicorrelated blocks of 50 with within-block correlation $\rho = 0.3$ by construction (empirically 0.265 in the realized draw), mimicking gene modules. Feature distributions are mixed: 30% of columns are pushed to heavy-tailed (t_3) or right-skewed (log-normal) marginals through the probability integral transform, which changes the marginals while preserving the block dependence structure, reflecting heterogeneous gene expression behavior.

s1 makes 20% of the informative features act through a nonlinear link and s2 makes 80%, cycling through quadratic, logarithmic, saturating (tanh), and threshold effects. s1 contains 5 pairwise interaction terms $X_i \times X_j$ drawn from the informative set, while s2 contains 10. Noise is Gaussian with standard deviation set to a fixed multiple of the signal standard deviation: 0.35 for s1, giving a realized signal-to-noise ratio of 8.48, and 0.50 for s2, giving 4.42. Each dataset was split 80/20 into train and test sets, stratified by response quantiles, to permit both feature-recovery evaluation on the training split and held-out predictive evaluation on the test split. The known ground truth enables direct assessment of feature recovery through TPR and FDR.

2.3.2 Real Curated Cancer Datasets from CuMiDa

Leukemia_GSE28497 is an imbalanced multiclass classification dataset whose target contains 7 leukemia subtypes (B-CELL_ALL, B-CELL_ALL_TCF3-PBX1, B-CELL_ALL_HYPERDIP, B-CELL_ALL_HYPO, B-CELL_ALL_MLL, B-CELL_ALL_T-ALL, B-CELL_ALL_ETV6-RUNX1), with 281 samples and 22,284 gene probes. It requires identification of subtle gene expression differences across imbalanced subtypes.

Liver_GSE14520_U133A is a binary classification dataset (hepatocellular carcinoma versus normal) with 357 samples and 22,278 gene probes.

Preprocessing. Both real datasets underwent duplicate-probe removal, median imputation of any missing values, removal of zero-variance features, StandardScaler normalization (zero mean, unit variance) fitted on the training split only, and a 70/30 train/test split with stratification. CuMiDa datasets were already extensively curated with background correction, normalization, and quality control [29].

2.4 Evaluation Framework

2.4.1 Two-Stage Selection and Prediction Pipeline

All methods were evaluated using a consistent two-stage pipeline that separates feature selection from downstream prediction.

Stage 1 (feature selection). Each method M receives $(X_{\text{train}}, \mathbf{y}_{\text{train}})$ and outputs a selected subset S_M. Methods use recommended literature parameters to avoid overfitting the selection step to the datasets.

Stage 2 (prediction evaluation). A LightGBM model is trained on $X_{\text{train}}[:, S_M]$ with hyperparameter tuning under 5-fold cross-validation, then evaluated on the held-out test set $X_{\text{test}}[:, S_M]$. The search space is

- Tree structure: `n_estimators` $\in \{50, 100, 200\}$, `max_depth` $\in \{3, 5, 7, 10, -1\}$, `num_leaves` $\in \{15, 31, 63, 127\}$
- Learning dynamics: `learning_rate` $\in \{0.01, 0.1, 0.2, 0.5, 0.7, 1\}$, `min_child_samples` $\in \{10, 20, 30, 50\}$
- Regularization: `subsample` $\in \{0.6, \ldots, 1.0\}$, `colsample_bytree` $\in \{0.6, \ldots, 1.0\}$, `reg_alpha` $\in \{0.01, 0.1, 0.2, 0.5, 0.7, 1.0\}$, `reg_lambda` $\in \{0, 0.01, 0.1, 0.2, 0.5, 0.7, 1.0\}$

Exhaustive enumeration of this space is roughly 1.5×10^6 configurations, which at 5 folds would require on the order of 10^7 model fits per method and is not tractable. We therefore use randomized search over the same space with 60 sampled configurations per method, drawn under a fixed seed so that every method sees the identical budget and the identical candidate stream. Optimization targets R^2 for regression, AUC-ROC for binary classification, and macro F1 for multiclass.

All methods are judged by the same downstream predictor, isolating feature selection quality. LightGBM is chosen for its performance on tabular data, computational efficiency, and ability to handle nonlinear relationships [30].

2.4.2 Replication and Uncertainty

Every configuration is replicated across five seeds. A seed controls the train/test split and the internal randomization of every selector, so a replicate is a full end-to-end repetition rather than a re-initialization of one component. The underlying data is held fixed across replicates, which keeps the same feature universe on offer in every run and is what makes the stability measurement below well defined; on the real datasets this is exactly the resampling protocol the data permits, and applying it to the synthetic datasets as well keeps all four benchmarks comparable.

Every reported quantity is the mean over replicates, and every error bar is the t-based 95% confidence interval on that mean. With five replicates these intervals are wide, and they are meant to be read as such: a difference between two methods whose intervals overlap is not evidence of a real difference, and several of the gaps reported below fall into that category. This is a deliberate correction to the single-run design of the earlier draft, where apparent rankings could not be distinguished from seed noise.

Replication also permits a direct measurement of the property this literature calls stability but rarely quantifies. For each method we compute the mean pairwise Jaccard index between its own selected sets across the replicate seeds,

$$\text{Stab}(M) = \binom{R}{2}^{-1} \sum_{r<r'} \frac{|S_M^{(r)} \cap S_M^{(r')}|}{|S_M^{(r)} \cup S_M^{(r')}|}, \tag{8}$$

over R replicates. A value of 1 means the method returns the same feature set every time; a value near 0 means the returned signature is essentially a fresh draw on each run, which would undermine any biological interpretation of it.

2.4.3 Performance Metrics

For predictive performance we report R^2, RMSE and MAE for regression; accuracy, F1, precision, recall and AUC-ROC for binary classification; and accuracy, macro F1 and weighted F1 for multiclass. Feature recovery on the synthetic datasets is measured through TPR, FDR, precision and selection F1. Computational efficiency is the wall-clock feature selection runtime in minutes. Sparsity is the number of features selected, $|S|$, and the sparsity ratio $|S|/p$. Cross-method agreement is measured by the Jaccard similarity $|S_i \cap S_j|/|S_i \cup S_j|$ and the overlap coefficient $|S_i \cap S_j|/\min(|S_i|, |S_j|)$.

All of these are reported as a mean over the replicate seeds with a confidence interval, per Section 2.4.2. This evaluation framework enables assessment of the central trade-offs in feature selection: predictive performance against sparsity, sensitivity against specificity in feature recovery, stability against flexibility, and computational cost against selection quality.

2.5 Reproducibility

All methods, data generators, the experiment driver, and the figure and table generation code are available at `https://github.com/KashfiR/SPARCS`. Every result reported below is produced by `experiments.py`, which writes a machine-readable record of the selected feature sets, metrics, runtimes, per-replicate values, and the exact parameter configuration used; `figures.py` rebuilds every figure and Table 1 from that record, so no reported number is transcribed by hand. Each replicate is checkpointed individually and is identified by a fingerprint of the settings that produced it, so a run can be spread over several sessions and additional seeds can be added later without recomputing the existing ones. The synthetic datasets are generated from fixed seeds and require no download.

3 Results and Discussion

Table 1: Method performance across the four benchmark datasets.

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3.1 Synthetic Dataset Performance and Feature Recovery

3.1.1 Moderately Nonlinear Data (s1)

On the synthetic dataset with 20% nonlinearity, SPARCS achieved the highest predictive performance and the highest True Positive Rate of the four methods, recovering the largest share of truly informative features. All methods nonetheless exhibited high False Discovery Rates, reflecting the difficulty of isolating truly informative features in high-dimensional settings. CPSS-Spearman showed the lowest FDR by selecting very few features in total, but this extreme conservatism came at the cost of missing the overwhelming majority of true features. SPARCS, ASGDR and Spearman-ISIS selected at or near the feature budget with similar FDRs, indicating comparable specificity at matched selection size.

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Figure 3: Feature recovery (TPR and FDR) on the two synthetic datasets.

Figure 1 and the recovery panels together illustrate that SPARCS and ASGDR achieved the best balance between TPR and FDR. CPSS-Spearman's very low TPR highlights the trade-off between stability and sensitivity: while CPSS provides formal error control, it may be overly conservative for exploratory analysis.

3.1.2 Highly Nonlinear Data (s2)

The second synthetic dataset is intentionally more difficult: 80% nonlinear informative features, twice as many interaction terms, higher noise, and a lower signal-to-noise ratio. This design tests method robustness under severe complexity.

Performance patterns revealed a striking finding. ASGDR improved as complexity increased, while all screening-based methods deteriorated; SPARCS declined and Spearman-ISIS collapsed, performing worst despite being competitive on the easier dataset. CPSS-Spearman maintained moderate performance.

This counterintuitive result, where adaptive regularization improves while screening methods fail, reveals that when nonlinearity, interactions, and noise become pervasive, linear methods with strong regularization may be more robust than sophisticated nonlinear screening. ASGDR's L_1 and L_2 penalties enforce sparsity and stability simultaneously, preventing overfitting to spurious nonlinear patterns. In contrast, screening methods attempt to detect marginal and conditional associations that become increasingly unreliable as noise dominates signal. The interaction terms further confound marginal screening, since features important only through an interaction appear marginally weak and are discarded.

Feature recovery deteriorated across the board in this scenario. ASGDR achieved the highest TPR, CPSS-Spearman again selected very few features with minimal recovery, and False Discovery Rates increased for every method. The comparable TPR between SPARCS and Spearman-ISIS suggests that RDC-based nonlinear screening was insufficient to overcome the dominance of highly nonlinear effects, though SPARCS maintained better predictive performance than Spearman-ISIS.

3.1.3 Computational Efficiency on Synthetic Data

Runtime analysis reveals stark differences in computational efficiency. ASGD is consistently fastest, leveraging efficient SGD optimization. Spearman-ISIS is also rapid through fast marginal screening. SPARCS requires moderate time due to RDC computation, permutation testing, and batch CPSS. CPSS-Spearman is slowest, requiring 60 applications of the base selector (30 pairs $\times$ 2 halves). Despite being substantially slower than Spearman-ISIS, SPARCS delivers better predictive performance and feature recovery on the moderately nonlinear dataset, a favorable trade-off between computation and selection quality for discovery work.

3.2 Real Cancer Dataset Performance

3.2.1 Binary Classification: Liver Cancer

For binary classification of hepatocellular carcinoma versus normal liver tissue (357 samples, 22,278 probes), all four methods achieved excellent predictive performance, demonstrating the viability of the selected feature sets for clinical classification. SPARCS, Spearman-ISIS and CPSS-Spearman achieved effectively identical accuracy, F1 and recall, with near-identical precision; Spearman-ISIS achieved a marginally higher AUC. ASGDC showed slightly lower but still strong performance.

The similar performance across methods suggests that for this well-separated binary classification problem, multiple feature subsets can achieve high accuracy. Three of the four methods converged on the same selection size, with CPSS-Spearman slightly larger, indicating broad agreement on the appropriate sparsity level for this dataset.

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Figure 4: Test-set performance against feature selection runtime.

The performance-runtime panel shows ASGDC achieving comparable performance at a small fraction of the cost, while CPSS-Spearman is by far the slowest, with SPARCS and Spearman-ISIS between the two extremes. For production deployment ASGDC's speed advantage is substantial; for research requiring rigorous error control, CPSS-Spearman's runtime may be acceptable.

3.2.2 Multiclass Classification: Leukemia Subtype Discrimination

The 7-class leukemia subtype task (281 samples, 22,284 probes) separated the methods more clearly than the binary case. ASGDC achieved the highest accuracy and macro F1, substantially outperforming the others. SPARCS achieved the second-best macro F1, with Spearman-ISIS and CPSS-Spearman performing similarly but slightly worse.

Notably, SPARCS terminated on its own stopping criterion well below the feature budget, selecting roughly half as many features as the other methods while remaining competitive. This sparsity is a consequence of the adaptive threshold and stopping rule of Section 2.2.1: selection ends when no candidate clears both the stability and RDC magnitude thresholds, rather than continuing until a fixed target is met.

ASGDC's stronger performance while selecting more features than SPARCS indicates that adaptive regularization may be particularly effective for multiclass problems, since the L_2 penalty stabilizes coefficient estimation across multiple classes, though it is less useful where high sparsity and human interpretability are the priority. ASGDC was also the fastest method, making it attractive for multiclass applications where computational efficiency is critical.

3.3 Method Stability and Feature Set Overlap

3.3.1 Cross-Method Agreement Patterns

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Figure 5: Jaccard similarity heatmaps quantifying the overlap between every pair of methods.

Spearman-ISIS and CPSS-Spearman show the highest agreement on the real datasets but markedly lower agreement on the synthetic ones. Strong agreement on real data is expected, since CPSS-Spearman uses Spearman-ISIS as its base selector and retains the features it selects most frequently; the high overlap confirms that CPSS is identifying the stable core of ISIS's selections. The weaker agreement on synthetic data suggests either that the synthetic response lacks the shared structure that unifies the two methods on real data, or that the synthetic dependence patterns are simpler than their biological counterparts. Either way, the divergence is a caution against reading synthetic feature-recovery numbers as a proxy for real-data behavior.

ASGD shows minimal overlap with screening-based methods on the real datasets , with substantially higher overlap on synthetic data. This low overlap indicates that adaptive regularization and iterative screening identify fundamentally different feature subsets, and again suggests that the synthetic data may not be representative. ASGD's linear optimization prioritizes features with strong marginal and conditional linear effects, while screening methods, and especially SPARCS with RDC, can detect features with weaker linear but stronger nonlinear associations.

SPARCS shows moderate agreement with both paradigms. Its intermediate position reflects its hybrid nature: it incorporates elements of both screening (ISIS-style iteration) and stability (CPSS) while adding nonlinear detection (RDC). The moderate overlap suggests SPARCS identifies a partially distinct feature set, potentially capturing nonlinear features missed by purely linear or purely marginal methods.

Overlap decreases with increasing nonlinearity. Comparing the 20% and 80% nonlinear synthetic datasets, Jaccard indices fall for most method pairs. Methods diverge more when nonlinear effects dominate, as different approaches prioritize different aspects of the feature-response relationship.

3.3.2 Detailed Feature Intersection Analysis

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Figure 6: Number of features shared by each pair of methods across the four datasets.

The SPARCS and Spearman-ISIS intersection is moderate throughout, indicating that although the two share a rank-based screening foundation, the additional SPARCS components (RDC, permutation testing, batch CPSS) shift the selected set considerably: partial methodological alignment with distinct selection priorities.

The SPARCS and CPSS-Spearman intersection is highly dataset-dependent. On the synthetic datasets the intersection is limited by CPSS's extreme conservatism, so it accounts for nearly all of CPSS's selections but only a small fraction of SPARCS's. On the real datasets overlap increases substantially, suggesting better agreement when the signal-to-noise ratio is higher. The overlap is asymmetric because the two methods select sets of very different size.

Multiclass classification showed the most distinct pattern, with SPARCS selecting a small set that is largely contained in the larger sets chosen by the two screening methods. This high containment suggests SPARCS identified a core subset of features that the other methods also recognized as important, achieving comparable accuracy with superior parsimony. The very low SPARCS-ASGDC intersection points to fundamentally distinct feature priorities between integrated screening and adaptive regularization.

3.3.3 Selection Stability Under Reseeding

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Figure 7: Selection stability: the mean pairwise Jaccard index between a method's own selected sets across replicate seeds. Higher is more reproducible.

Cross-method overlap answers whether two methods agree with each other; it does not answer whether a single method agrees with itself. The replicate design lets us measure the latter directly, and it is the more consequential quantity for biomarker work, since a signature that changes when only the split changes cannot carry a biological interpretation regardless of how well it predicts. The stability values should be read alongside the confidence intervals in Table 1: a method can be both accurate on average and unstable in what it returns, and those are separate failure modes with separate consequences.

3.3.4 Implications for Method Selection Stability

The stability analysis reveals a critical trade-off. **CPSS-Spearman provides maximal stability** by construction, having the highest self-agreement across subsamples, but **sacrifices sensitivity**: very few features are selected and TPR on synthetic data is low. **SPARCS balances stability and performance**, incorporating CPSS-based stability assessment while using adaptive thresholds and batch addition to maintain reasonable sensitivity. Its moderate Jaccard indices place it neither at the maximally stable end (like CPSS) nor at the maximally flexible end (like pure screening), a middle ground that may be well suited to exploratory biomarker discovery.

The low cross-paradigm agreement between screening and regularization raises important questions about feature selection consensus. When different valid methods select largely non-overlapping feature sets yet achieve similar predictive performance, it suggests either that multiple equivalent feature sets exist (redundancy in the feature space) or that different methods capture complementary aspects of the feature-response relationship. The consistent performance of diverse feature sets in binary classification supports the redundancy hypothesis for that dataset, while the performance differences in multiclass classification suggest complementary information.

3.4 Sparsity and Computational Trade-offs

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Figure 8: Selection sparsity and runtime across methods and datasets. Runtime is on a logarithmic scale.

CPSS-Spearman exhibits extreme variability in sparsity between the synthetic and real datasets. This instability stems from its selection frequency threshold: when the signal is weak, few features exceed τ, whereas when the signal is strong many features appear stable. This dataset-dependent behavior limits CPSS's utility when a desired sparsity level is predetermined.

SPARCS shows more consistent sparsity, with the notable exception of multiclass classification where the adaptive threshold identified a natural stopping point well below the budget. This data-driven sparsity is desirable: SPARCS adapts to the structure of each dataset rather than imposing a fixed feature count. Spearman-ISIS and ASGD select up to their budget by construction, giving predictable sparsity but potentially missing dataset-specific optimal sizes.

Runtime differences span orders of magnitude, with ASGD fastest, then Spearman-ISIS, then SPARCS, then CPSS-Spearman. The gap between the fastest and slowest methods is close to three orders of magnitude and has practical consequences: for routine clinical deployment ASGD's near-instantaneous selection is highly attractive, while for one-time biomarker discovery studies SPARCS's runtime is acceptable given its stronger feature recovery and balanced performance.

3.5 SPARCS: Strengths and Limitations

SPARCS integrates rank-based SIS, RDC residual screening, permutation testing, and batch CPSS. This multilayered design improved feature recovery in moderately complex settings, achieving the highest TPR alongside competitive predictive performance. It balances sensitivity and specificity better than CPSS, which is overly conservative, by using adaptive thresholding to set sparsity from the data rather than from a fixed target. Its runtime is acceptable for discovery studies where improved recovery outweighs the extra compute.

SPARCS struggles under extreme nonlinearity: on the 80% nonlinear dataset it produced negative R^2 and low recovery, whereas ASGDR improved. RDC can miss complex interactions or threshold effects, and its bandwidth s trades sensitivity to high-frequency dependence against the level of the null. The computational overhead of RDC, permutations and CPSS is only justified when it finds signal that simpler methods miss; on easy, well-separated problems simpler methods suffice. Multiple hyperparameters (M_{pre}, M_{rdc}, n_{perm}, α, γ, k_{add}, the RDC threshold) affect performance, and theoretical guarantees remain limited compared with CPSS. SPARCS can also be overkill for straightforward binary classification.

4 Discussion

4.1 Main Paradigm Takeaways

Adaptive regularization (ASGD) is the fastest method and the most robust in highly complex and noisy settings. It favors features with strong linear contributions and has low overlap with screening methods. Greedy iterative screening (Spearman-ISIS) provides an excellent speed-accuracy balance and an interpretable selection path, but lacks any stability assessment. Stability selection (CPSS-Spearman) guarantees maximal stability with theoretical backing, but can be too conservative for discovery in weak-signal settings. SPARCS, integrating screening and stability, occupies a middle ground: better recovery than pure screening at moderate complexity, more sensitive than CPSS, but not always superior under pervasive nonlinearity. SPARCS is most useful for exploratory biomarker discovery in noisy or nonlinear problems where sensitivity matters and moderate runtimes are acceptable.

Low cross-paradigm overlap implies that multiple equally predictive but different feature sets exist, reflecting either redundancy or complementary signal. For robust biomarkers, consensus or ensemble approaches, such as the intersection of SPARCS and ASGDR, or their union filtered by CPSS stability, would prioritize features with higher reproducibility.

Unlike CPSS-Spearman, which achieves low FDR through extreme conservatism, SPARCS maintains moderate FDR while selecting substantially more features. This balance matters for exploratory biomarker discovery, where missing true features is more costly than including false positives that can be filtered in subsequent validation pipelines.

4.2 Limitations

Three limitations bound the conclusions. First, the synthetic datasets differ markedly from the real omics datasets in the cross-method overlap they induce, so synthetic feature-recovery numbers should be read as a controlled diagnostic rather than a prediction of real-data behavior. Second, five replicates is enough to expose seed sensitivity but not enough to estimate it precisely; the confidence intervals are correspondingly wide, and differences smaller than those intervals should not be read as rankings. A larger replicate budget, or a paired test across seeds, would sharpen the comparisons that currently overlap. Third, the comparison covers two real datasets from a single curated repository and one downstream predictor, so generalization to other assay types, disease areas, and model classes is untested.

4.3 Future Work

Broader benchmarking across more datasets and against tree-based, deep and Bayesian selectors is needed. In particular, classification with more realistic synthetic omics data, specifically using Model-X knockoffs, should be tested, and extensive hyperparameter tuning of SPARCS's parameters will be pursued [31]. Alternative dependence measures such as the Hilbert-Schmidt independence criterion and the computationally intensive distance correlation, artificial neural network screening, adaptive hyperparameter selection, and theoretical analysis of SPARCS for Type I error control and sure-screening conditions will also be explored [32, 33]. Integrating uncertainty quantification such as bootstrap confidence intervals or Bayesian posteriors would improve interpretability and clinical translation [34]. Increasing the replicate budget beyond five seeds, and adding paired significance testing across seeds, is the most immediate priority for the comparisons whose intervals currently overlap.

5 Conclusions

This paper contributes an empirical, systematic comparison of four paradigms (the novel SPARCS, Spearman-ISIS, CPSS-Spearman, and ASGD) across synthetic and real cancer genomics datasets to highlight trade-offs in feature recovery, stability, runtime, and parsimony. Feature selection remains a central challenge in genomic medicine, as robust biomarker identification is essential for precision diagnostics and targeted therapies. Our results show that no single method is universally optimal; performance depends on dataset characteristics, computational constraints, and application goals, with all current approaches achieving only moderate success in complex, nonlinear settings. Future advances are likely to arise from integrating biological prior knowledge, developing methods tailored to nonlinear and interaction-rich signals including deep learning approaches, and using ensemble strategies to capture complementary information. Within this landscape, SPARCS represents a step toward integrative feature selection, enabling more reliable and interpretable biomarker discovery that can improve clinical decision-making, enhance diagnostic and prognostic prediction, accelerate drug development, and reduce healthcare costs through smaller, more precise molecular panels.

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